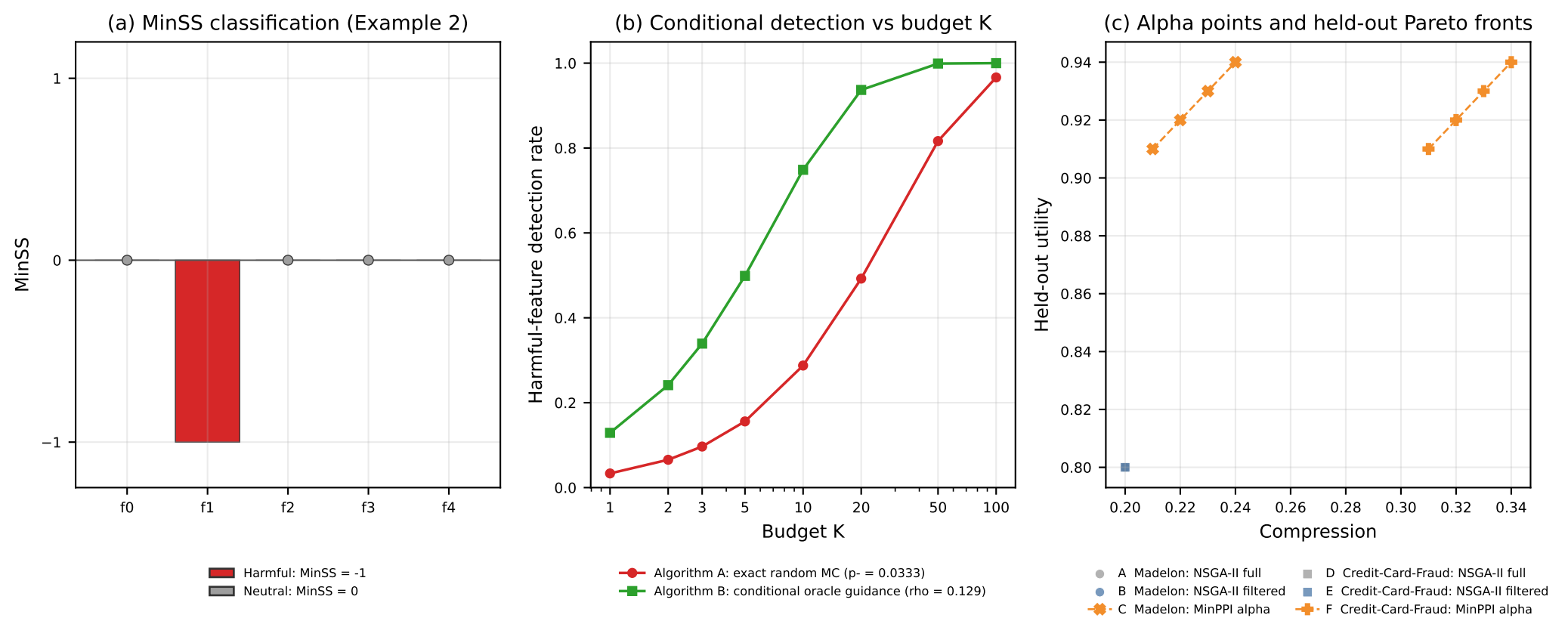
(a) MinSS classification (Example 2)
MinSS
1
0
−1
f0
f1
f2
f3
f4
Harmful: MinSS = -1
Neutral: MinSS = 0
(b) Conditional detection vs budget K
Harmful-feature detection rate
0.0
0.2
0.4
0.6
0.8
1.0
1
2
3
5
10
20
50
100
Budget K
Algorithm A: exact random MC (p- = 0.0333)
Algorithm B: conditional oracle guidance (rho = 0.129)
(c) Alpha points and held-out Pareto fronts
Held-out utility
0.80
0.82
0.84
0.86
0.88
0.90
0.92
0.94
0.20
0.22
0.24
0.26
0.28
0.30
0.32
0.34
Compression
A Madelon: NSGA-II full
B Madelon: NSGA-II filtered
C Madelon: MinPPI alpha
D Credit-Card-Fraud: NSGA-II full
E Credit-Card-Fraud: NSGA-II filtered
F Credit-Card-Fraud: MinPPI alpha